

Handwritten Digit Classification Using TensorFlow - Keras :-

① Introduction :-

Artificial Intelligence (AI) and Machine Learning (ML) are widely used for solving real-world problems such as image recognition, speech processing, and recommendation systems.

One of the most important applications of AI is image classification, where a computer learns to identify objects present in an image.

In this assignment, a neural network is developed using TensorFlow/Keras to classify handwritten digits from 0 to 9. The MNIST dataset is used because it is a standard benchmark dataset for learning and evaluating image classification models. The objective is to train a model that can correctly recognize handwritten digits with high accuracy.

② Dataset Details :-

The MNIST (Modified National Institution of Standards and Technology) dataset contains grayscale images of handwritten digits. It is one of the most commonly used datasets in machine learning and deep learning research.

Details :-

→ Training Images : 60,000

→ Testing Images : 10,000

→ Image Size : 28 x 28 pixels

→ Number of Classes : 10 (digits 0-9)

→ Image Type : Grayscale

③ Methodology :-

The implementation was carried out using python and TensorFlow/Keras.

① Data Loading :-

The MNIST dataset was loaded directly from the TensorFlow Keras library. The dataset was automatically divided into training and testing sets.

② Data Preprocessing :-

The pixel values of all images were divided by 255 to normalize them between 0 and 1. Normalization helps the neural network train faster and improves numerical stability.

③ Model Design :-

A simple feed-forward neural network was created with the following architecture:

Input Image (28×28)

↓ Flatten Layer

Dense Layer (128, neurons, ReLU)

↓ Dense Layer (64 neurons, ReLU)

↓ Output Layer (10 neurons, Softmax)

④ Model Training :-

The model was compiled using:

→ Optimizer: Adam

→ Loss Function: Sparse Categorical Crossentropy

→ Evaluation Metric: Accuracy

④ Results :-

- Test Accuracy: $\sim 98\%$
- Correctly classified most digits
- Accuracy and loss graphs generated
- 5 sample predictions tested

The model correctly classified most of the selected handwritten digits, demonstrating its ability to generalize to unseen data.

⑤ Experiment :-

Added Dropout layer.

Result :-

- Reduce overfitting
- Slight improvement in accuracy

Observation :-

The model with Dropout achieved slightly higher accuracy and showed better generalization performance. The addition of Dropout reduced the possibility of overfitting and improved the robustness of the neural network.

⑥ Conclusion :-

The neural network successfully classified handwritten digits with high accuracy and demonstrated the effectiveness of deep learning for image classification.